

Derivatives of Inverse Functions

Date _____ Period _____

For each problem, find $(f^{-1})'(x)$ by direct computation.

1) $f(x) = -3x + 3$

$$(f^{-1})'(x) = -\frac{1}{3}$$

2) $f(x) = -2x + 3$

$$(f^{-1})'(x) = -\frac{1}{2}$$

For each problem, find $(f^{-1})'(x)$ by using the theorem $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

3) $f(x) = -5x + 1$

$$(f^{-1})'(x) = -\frac{1}{5}$$

4) $f(x) = -2x + 2$

$$(f^{-1})'(x) = -\frac{1}{2}$$

5) $f(x) = \sqrt{-2x - 3}$

$$(f^{-1})'(x) = -\frac{2x}{2}$$

6) $f(x) = -4x^3 - 4$

$$(f^{-1})'(x) = -\frac{1}{12 \cdot \left(\frac{-x-4}{4}\right)^{\frac{2}{3}}}$$

For each problem, find $(f^{-1})'(x)$ by using the formula $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$, where $y = f^{-1}(x)$

7) $f(x) = x^7 + x - 3$

$$(f^{-1})'(x) = \frac{1}{7y^6 + 1}$$

8) $f(x) = 3x^5 + 2x + 5$

$$(f^{-1})'(x) = \frac{1}{15y^4 + 2}$$